ABEL FARES

Engineering student — Systems & performance-oriented software

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PROFILE

Engineering student at ENSEEIHT with a strong background in mathematics and scientific reasoning (CPGE MPSI/MP, Lycée Pierre-de-Fermat). Primary interest in **systems and performance-oriented software engineering** (graphics, rendering, computation-heavy pipelines). Coherent secondary track toward **applied mathematics, scientific computing, and imaging**.

EDUCATION

ENSEEIHT (Toulouse INP)

Engineering degree — Computer Science / Applied Systems

2024 – Present Toulouse

- Coursework increasingly focused on image processing, mathematics, and computational systems.

Lycée Pierre-de-Fermat

CPGE MPSI / MP

2022 – 2024 Toulouse

- Rigorous training in mathematics and physics; problem-solving under constraints.

Baccalauréat général

Mathematics & Physics

2022 France

EXPERIENCE

INRAE

Research internship — Data analysis

Jul 2025 – Aug 2025 France

- Data preparation and analysis in a research environment (Python/pandas).
- Exposure to statistical modeling, including Bayesian regression approaches.

Private tutoring

Mathematics / Physics

Oct 2024 – Present Toulouse

- Pedagogy, communication, and structuring explanations for different levels.

PROJECTS

Voxel Engine

Java / OpenGL — Real-time rendering & systems experimentation

2024 – Present

- Chunk-based world representation, procedural terrain generation, rendering pipeline optimisation.
- Multi-threaded generation decoupling computation from rendering to avoid frame drops.

Mini-shell

C — Processes, signals, pipes

2024 – 2025

- Linux mini-shell with basic job control, signal handling, and simple pipelines.

Dijkstra (directed graph)

C — Algorithms & data structures

2024 – 2025

- Implementation using linked structures; distances, predecessors, path reconstruction.

Scientific computing — Eigenvalues

MATLAB

2024 – 2025

- Numerical methods for eigenvalue computation; convergence and stability analysis.

Image processing & classification

Python

2025

- Edge detection pipelines and basic classification approaches.

SKILLS (EACH LINKED TO A PROJECT)

- **Java** — Voxel Engine (architecture, rendering pipeline, performance constraints).
- **C** — Mini-shell (processes, signals, pipes) | Dijkstra (graphs, shortest paths).
- **MATLAB** — Scientific computing (eigenvalue algorithms, convergence analysis).
- **Python (ML)** — Google Colab: TensorFlow/Keras, scikit-learn (CNNs, neural networks).
- **Python (Data)** — pandas at INRAE; Bayesian regression workflows.
- **Image processing** — edge detection and basic classification pipelines.
- **Tools** — Git; debugging and profiling mindset (measure → optimise).

INTERESTS

Strength training, running and hiking, reading, drawing, cooking.